

Muhammad Ravi Shulthan Habibi

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SUMMARY

AI/ML Engineer and Computer Science graduate (Cum Laude, University of Indonesia) with expertise in NLP, large language models, and applied machine learning. Experienced in deploying AI systems for finance automation, computer vision, and conversational AI, with recognition as a Rising Star at Mekari and finalist in the 2024 ML Olympiad. Published at ACL & EMNLP and certified across Azure, GCP, and AWS; active contributor to IndoNLP and SEACrowd communities. Strong foundation in Natural Sciences, reaching provincial-level recognition in the 2012 National Science Olympiad.

WORK EXPERIENCE

Associate AI/ML Engineer & Data Scientist [Mekari](#) | Feb 2024 - Present

- Recognized as **Rising Star Q2 2025** for rapid impact, initiative, and ownership by company leadership.
- Developed an **Agentic AI system** for knowledge-base and database interaction, reaching up to **100% accuracy on Indonesian queries** and **~97% on English queries** using a state-of-the-art LLM.
- Deployed **production-ready systems for document verification: YOLO-based signature/stamp detection and CLIP-based spoof detection**, reducing annual costs by **~60%** while maintaining high accuracy (up to **99%**).
- Built **AI solutions for finance automation**, including a **Text2SQL-powered chatbot** for natural language queries, a **prompt-based invoice generator**, and **reconciliation features**.
- Designed and deployed an **automated document tagging system (~90% accuracy)** and an **invoice/receipt extraction model (F1 ~92%)** to streamline data processing.
- Led development of a **voice transcription system** with state-of-the-art results (**WER 1.84%, CER 1.90%**), improving accuracy for multilingual use cases.
- Enhanced the company's **AI chatbot** with intent classification, knowledge-base integration, and function calling, reducing response latency by **~66%**.
- Conducted **large-scale experiments on 50+ LLMs and encoder-only models** for NER and function calling, identifying **cost-effective deployment alternatives**.
- Delivered **biweekly knowledge-sharing sessions**, mentoring peers on **ML best practices**, and **RAG knowledge base design**.

AI/NLP Researcher Collaborator [University of Indonesia](#) | Aug 2023 - Present

- Collaborated with **IndoNLP** and **SEACrowd** on **multilingual and multimodal NLP research** with international partners.
- Co-authored **papers accepted at top-tier NLP venues (EMNLP 2024, ACL 2025)**; additional manuscripts **under review**.
- Led **dataset pipeline development** and **multilingual evaluation** for benchmarking, contributing to **large-scale experimentation, model analysis, and academic writing**.

Machine Learning Advisor [Bangkit Academy](#) | May 2024 - July 2024

- Mentored **2 teams (14 mentees)** on **ML capstone projects**; **mentee satisfaction: 5/5**.

Teaching Assistant of Advanced Data [University of Indonesia](#) | Jan 2023 - July 2023

- Taught **30+ students** on **DB optimization, data warehousing, big data, and distributed DB**.

PROJECTS

Understanding How Aya Thinks (Cohere Aya Expedition)	Repository
Analyzed multilingual behavior of Aya LLM (23 languages), uncovering English-first bias and low cross-lingual factual consistency (~34.5%).	
Earthquake Damage Classification (ML Olympiad 2024)	Repository
Finalist; ranked 3rd out of 84 participants with ensemble ML models, achieving top leaderboard score via feature selection & hyperparameter tuning.	
Hand Sign Image Classification (Backdrop Labs)	Repository
Built CNN achieving ~98% accuracy; deployed as proof-of-concept accessibility tool.	
Indonesian QA System using Natural Language Inference (Bachelor Thesis)	Details
Developed an Indonesian QA system using NLI, improving F1 by 5.7 points and analyzing context-question-answer types that benefit most.	

SKILLS

Programming:	Python, SQL, R.
AI/ML:	Large Language Models (RAG, fine-tuning, evaluation, function calling), NLP, Computer Vision, Deep Learning, Classical ML.
Technologies:	PyTorch, TensorFlow, Hugging Face, Scikit-Learn, Keras, OpenCV, LangChain, CLIP.
MLOps:	Model deployment with Docker & Kubernetes, CI/CD, monitoring & experiment tracking (TensorBoard, MLflow, Weights & Biases), SageMaker.
Cloud/Platforms:	GCP, AWS, Azure, FastAPI, Git.

CERTIFICATIONS

Microsoft:	Azure AI Engineer & Azure Data Scientist.
Google:	Associate Cloud Engineer & Professional Data Analytics.
AWS:	Cloud Essentials.
Harvard University:	CS50AI, CS50P, CS50X.
Other:	Oracle, Cisco, Kaggle.

EDUCATION

2019 - 2023	Bachelor's Degree (Computer Science) at University of Indonesia	(Cum Laude)
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PUBLICATIONS

- Lopo, J. A., **Habibi, M. R. S.** et al. (2025). *Language Surgery in Multilingual Large Language Models*. arXiv preprint arXiv:2506.12450.
- Cahyawijaya, S., **Habibi, M. R. S.** et al. (2025). *Crowdsourced, Crawl, or Generate? Creating SEA-VL, a Multicultural Vision-Language Dataset for Southeast Asia*. ACL 2025.
- Lovenia, H., **Habibi, M. R. S.** et al. (2024). *SEACrowd: A Multilingual Multimodal Data Hub and Benchmark Suite for Southeast Asian Languages*. EMNLP 2024.